

Assignment method comparison · MILP vs. alternatives for the inner-loop material matching problem

Method	Optimality guarantee	Hard constraint support	Inventory limits (RS $\times 1$, NS $\times \infty$)	Scalable to 120 slots	Computation time
Greedy <i>sequential, path-dependent</i>	✗	~	✓	✓	< 1 ms
Hungarian <i>1-to-1 optimal, no extra constraints</i>	~	✗	✗	✓	< 1 ms
GA <i>heuristic, no optimality proof</i>	✗	✗	~	~	seconds
RL <i>approximate, reward-engineering required</i>	✗	✗	✗	✗	minutes
MILP ★ <i>branch-and-bound, globally exact</i>	✓	✓	✓	✓	< 1 s

✓ fully supported ~ partial / limited ✗ not supported